

Syllabus

Parsons School of Design
Communication Design Department

Device Art

PSAM 2230 | CRN 3605 | Section A
Fall 2020
Wednesday, 7-9:40pm

[Ariel Churi](#)

[Zoom](#) Password: 482363

[Canvas](#)

[Device Art Discord server](#)

[Miro](#)

[are.na](#)

Course Description

Device Art is a platform for students to invent new works of art that do not distinguish the object from a tool, the mechanism from the concept; through playfulness these devices are caught between definitions of art, design, and engineering.

In this course devices are the artwork themselves; the mechanisms of the piece become part of the concept behind the work and the very essence of “making” and “play” is embedded in each of the pieces.

Device Art can break the mold of art commercialization, propelled by the new forms of manufacturing and the ease of prototyping as production tools become more accessible. These artworks are easily replicable and can become commercialized, and even display features of playful utility for everyday life.

Students will learn new methods for prototyping by combining new digital fabrication techniques as well as commonly used fabrication processes including re-purposing readymades. They will explore their own artistic concepts and develop a deeper understanding of how their ideas can be reproduced.

Device Art will challenge students to rethink their ideas of what art can become, and where the threshold lies between playfulness, utility, and conceptual inquiry.

They will develop basic electronics and physical computing skills, such as microcontroller programming and circuitry design.

They will engage in fabrication experiments to rethink and envision new devices that can facilitate invention from an artistic perspective, gaining new skills in device design, computing media, and device fabrication.

They will use physical computing methods to expand on the field of open source hardware, learn interaction design methods and rapid prototyping tools for 3D printing.

This course will provide platform to enhance the design process of material experimentation and propel functional and accessible models of device art.

Open to: All university undergraduate degree students.

Some seats have been reserved for BFA Design & Technology majors.

Learning Outcomes

Gain a critical understanding of current and emerging making technologies. Establish proficiency with methods and practices of modeling, sketching, and prototyping. Formalize the process of idea generation and testing using a design thinking tool set.

Enhance the communication and presentation skills of students.

Final Grade Calculation

1/3 Projects

1/3 Assignments

1/3 Participation

Course Outline

Week 1

Introductions

What is Device Art?

Assignment

How to use are.na

How to use Miro

Purchase materials

Week 2

Kick off Desisitive Technology.

3D modeling for 3D printing

Assignment

Unvirtual eMoji

Desistive Technology - Group Brainstorm

Week 3

Cardboard Prototyping

Electricity

Amps, Ohms, and Volts and how to test them.

Assignment

Desistive Technology - Cardboard prototype

Week 4

Testing

Electronics

Prototyping and circuit design

Basic circuits

Assignment

Test you prototype

Desistive Technology Video

Week 5

Share you *Desistive Technology* Video

Circuits
Sensing and using electricity

Assignment

Begin *Hactivism*
Design and create a circuit
Choose and order an integrated circuit

Week 6

Arduino
Coding

Assignment

Hello World
3D print *Unvirtual eMoji*

Week 7

Communicating with external integrated circuits

Assignment

Create your *Hactivism* video

Week 8

Share your *Hactivism* video

Assignment

IC show & tell

Week 9

Begin *Utopia/Dystopia*
PCB Design

Assignment

Create a PCB

Week 10

Casting & molding

Assignment

Duplicate *Unvirtual eMoji* or another object

Week 11

CAD Primer

Assignment

Utopia/Dystopia design documentation

Week 12

Electronic prototyping
Soldering

Assignment

Breadboard prototyping

Week 13

PCB soldering
Enclosure design

Week 14

Final check in *Utopia/Dystopia*

Week 15

Present *Utopia/Dystopia*

Assessable Tasks

The semester is divided into 3 design projects and a series of practical skills lessons with associated weekly assignments. Students will work in groups to design and critique their individual projects. Each hand-on skill lesson will be accompanied by a single week assignment.

Required Reading / Media

All assigned readings are in the *Files* section on *Canvas*.

Materials and Supplies

See the materials list and discussion on *Canvas*.

Class Resources

Resource links are in [are.na](#).

New School Resources

The university provides many resources to help students achieve academic and artistic excellence. These resources include:

[The University and associated Libraries](#)

[The University Learning Center](#)

For additional help with coursework assigned during the semester, you are encouraged to schedule tutoring sessions at the University Learning Center (ULC). Individual appointments are offered in Writing (all levels), Math, Adobe, Computer Programming, Oral Presentations and Time Management. Sessions are interactive, with both tutor and student participating. Appointments can be scheduled on Starfish or you can stop by for a walk-in session. The ULC is located on the 6th floor of 66 West 12th street. Academic and skill-building workshops are also offered. For a complete list of services, workshops, and general information, visit www.newschool.edu/learning

[center.](#)

University Disabilities Service

In keeping with the university's policy of providing equal access for students with disabilities, any student with a disability who needs academic accommodations is welcome to meet with me privately. All conversations will be kept confidential. Students requesting any accommodations will also need to contact Student Disability Service (SDS). SDS will conduct an intake and, if appropriate, the Director will provide an academic accommodation notification letter for you to bring to me. At that point, I will review the letter with you and discuss these accommodations in relation to this course.

Making Center

Faculty who are planning curriculum that makes use of specific resources should contact the Making Center in advance to coordinate.

The Making Center is a constellation of shops, labs, and open workspaces that are situated across the New

School to help students express their ideas in a variety of materials and methods. We have resources to help

support woodworking, metalworking, ceramics and pottery work, photography and film, textiles, printmaking,

3D printing, manual and CNC machining, and more. A staff of technicians and student workers provide expertise

and maintain the different shops and labs. Safety is a primary concern, so each area has policies for access,

training, and etiquette that students and faculty should be familiar with. Many areas require specific

orientations or trainings before access is granted. Detailed information about the resources available, as well as

schedules, trainings, and policies can be found at resources.parsons.edu.

Grading Standards

Undergraduate

A student's final grades and GPA are calculated using a 4.0 scale.

A [4.0]

Work of exceptional quality, which often goes beyond the stated goals of the course

A- [3.7]

Work of very high quality

B+ [3.3]

Work of high quality that indicates higher than average abilities

B [3.0]

Very good work that satisfies the goals of the course

B- [2.7]

Good work

C+ [2.3]

Above-average work

C [2.0]

Average work that indicates an understanding of the course material; passable
Satisfactory completion of a course is considered to be a grade of C or higher.

C- [1.7]

Passing work but below good academic standing

D [1.0]

Below-average work that indicates a student does not fully understand the assignments;
Probation level though passing for credit

F [0.0]

Failure, no credit

Graduate

A Work of exceptional quality

A- Work of high quality

B+ Very good work

B Good work; satisfies course requirements

Satisfactory completion of a course is considered to be a grade of B or higher.

B- Below-average work

C+ Less than adequate work

C Well below average work

C- Poor work; lowest possible passing grade

F Failure

GM Grade missing for an individual

Grades of D are not used in graduate level courses.

Grade of W

The grade of W may be issued by the Office of the Registrar to a student who officially withdraws from a course within the applicable deadline. There is no academic penalty, but the grade will appear on the student transcript.

Grade of Z

This grade is to be assigned to students who have never attended or stopped attending classes. Exceptions can be made if the student has completed enough work to warrant a grade (including a failing grade), and arrangements have been made with the instructor(s) and the Dean's Office prior to grade submission. The Z grade does not calculate into the student's GPA.

Grades of Incomplete

The grade of I, or temporary incomplete, may be granted to a student under unusual and extenuating circumstances, such as when the student's academic life is interrupted by a medical or personal emergency. This mark is not given automatically but only upon the student's request and at the discretion of the instructor. A Request for Incomplete form must be completed and signed by the student and instructor. The time allowed for completion of the work and removal of the "I" mark will be set by the instructor with the following limitations:

Undergraduate students: Work must be completed no later than the seventh week of the following fall semester for spring or summer term incompletes and no later than the seventh

week of the following spring semester for fall term incompletes. Grades of "I" not revised in the prescribed time will be recorded as a final grade of "F" by the Registrar's Office.

Graduate students: Work must be completed no later than one year following the end of the class. Grades of "I" not revised in the prescribed time will be recorded as a final grade of "N" by the Registrar's Office.

College, School, Program and Class Policies

Canvas

Use of Canvas may be an important resource for this class. Students should check it for announcements before coming to class each week.

Electronic Devices

The use of electronic devices (phones, tablets, laptops, cameras, etc.) is permitted when the device is being used in relation to the course's work. All other uses are prohibited in the classroom and devices should be turned off before class starts.

Responsibility

Students are responsible for all assignments, even if they are absent. Late assignments, failure to complete the assignments for class discussion and/or critique, and lack of preparedness for in class discussions, presentations and/or critiques will jeopardize your successful completion of this course.

Active Participation and Attendance

Class participation is an essential part of class and includes: keeping up with reading, assignments, projects, contributing meaningfully to class discussions, active participation in group work, and coming to class regularly and on time.

Parsons' attendance guidelines were developed to encourage students' success in all aspects of their academic programs. Full participation is essential to the successful completion of coursework and enhances the quality of the educational experience for all, particularly in courses where group work is integral; thus, Parsons promotes high levels of attendance. Students are expected to attend classes regularly and promptly and in compliance with the standards stated in this course syllabus.

While attendance is just one aspect of active participation, absence from a significant portion of class time may prevent the successful attainment of course objectives. A significant portion of class time is generally defined as the equivalent of three weeks, or 20%, of class time. Lateness or early departure from class may be recorded as one full absence. Students may be asked to withdraw from a course if habitual absenteeism or tardiness has a negative impact on the class environment.

I will assess each student's performance against all of the assessment criteria in determining your final grade.

Academic Honesty and Integrity

Compromising your academic integrity may lead to serious consequences, including (but not limited to) one or more of the following: failure of the assignment, failure of the course, academic warning, disciplinary probation, suspension from the university, or dismissal from the university.

Students are responsible for understanding the University's policy on academic honesty and integrity and must make use of proper citations of sources for writing papers, creating, presenting, and performing their work, taking examinations, and doing research. It is the responsibility of students to learn the procedures specific to their discipline for correctly and appropriately differentiating their own work from that of others. The full text of the policy, including adjudication procedures, is found at

<http://www.newschool.edu/policies/#> Resources regarding what plagiarism is and how to avoid it can be found on the Learning Center's website: <http://www.newschool.edu/university-learning-center/student-resources/>

The New School views "academic honesty and integrity" as the duty of every member of an academic community to claim authorship for his or her own work and only for that work, and to recognize the contributions of others accurately and completely. This obligation is fundamental to the integrity of intellectual debate, and creative and academic pursuits. Academic honesty and integrity includes accurate use of quotations, as well as appropriate and explicit citation of sources in instances of paraphrasing and describing ideas, or reporting on research findings or any aspect of the work of others (including that of faculty members and other students). Academic dishonesty results from infractions of this "accurate use". The standards of academic honesty and integrity, and citation of sources, apply to all forms of academic work, including submissions of drafts of final papers or projects. All members of the University community are expected to conduct themselves in accord with the standards of academic honesty and integrity. Please see the complete policy in the Parsons Catalog.

Intellectual Property Rights

The New School (the "university") seeks to encourage creativity and invention among its faculty members and students. In doing so, the University affirms its traditional commitment to the personal ownership by its faculty members and students of Intellectual Property Rights in works they create. The complete policy governing Intellectual Property Rights may be seen on the [university website, on the Provost's page](#).

Communication Design Policies

A. Communication Design Zero Tolerance Attendance Policy

In order to foster a studio learning environment where we all learn from peers and through dialogue, timely and regular attendance is a strict expectation for all Communication Design students. Students who are not present in class are unable to meet the learning outcomes of a Communication Design course.

For classes meeting once a week, students are allowed 2 absences. For classes meeting twice a week, students are allowed 4 absences. Any absence beyond the allowed absences will result in an automatic failure (F) for the course. There are no excused absences. This applies to each and every student.

A student is deemed tardy if a student fails to arrive within 15 minutes past the beginning of class. 2 tardies will result in an automatic absence. A student who arrives an hour past the beginning of class will be deemed absent.

B. No Late Work and Missed Critiques

Work that is submitted past the assignment due date will result in an automatic failure for the assignment. For work presented in critique, absence at the critique will result in an automatic failure for the assignment.

C. CD App

Information about upcoming CD Lectures, events from AIGA NY/TDC, the CD Library, and the CD Paper Store can be found on the CD App.

<https://cdparsons.glideapp.io/>

To install the app on your phone:

For iOS users:

1. Visit the link in Safari, tap on the "Share" icon located at the bottom of the page
2. Tap the "Add to Home Screen" button
3. Tap the "Add" button at the top right corner of the screen

For Android users:

1. Tap on the notification banner at the bottom of the page (alternatively, you can tap the "Add to Home Screen" option inside the menu at the top right corner of the screen)
2. Tap the "Add" button on the modal